TEAM NAME: 17_Ecolife

DOMAIN: Sustainability and Green Economy
PROBLEM STATEMENT: Resource Usage and Impact Analytics Platform

S.No	Team Members	Roll No
1	Kondaveeti Chetan Gana Sidhardha	S20240010118
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Functional Requirements

A. User Management & Authentication

- COO registers organization with name, industry type, and personal credentials
- All roles can log in, log out, and update profile details
- COO is the only role that can change email address
- Department Managers and Analysts receive login credentials via personal email upon account creation
- Role-based access control enforced — each role accesses only their designated features
- Forgot password supported via OTP verification

B. Organization & Staff Management

- COO creates organization profile, adds and manages departments with custom names
- COO adds, edits, and removes staff with name, email, mobile, designation, and platform access role
- System auto-sends login credentials to staff personal email on account creation
- COO monitors staff activation status and resends credentials if needed

C. Monthly Consumption Data Submission

- Department Manager selects reporting period using free month and year dropdowns
- Enters consumption data for default resources: Fuel, Diesel, LPG — custom types can be added or deleted
- Uploads invoice for auto-comparison against entered figures
- Smart validation checklist must fully pass before submission is allowed
- Report is locked on submission — no edits after locking
- System notifies Analyst when all departments have submitted

D. Targets & Alert Threshold Configuration

- COO sets individual monthly consumption target per resource type per department
- Custom resource types can be added or deleted when configuring targets
- COO sets alert threshold per department — alert triggers when exceeded
- Historical consumption chart auto-loads for reference before setting targets

E. Alerts & Escalation Management

- System triggers alert when department exceeds threshold — classified as Low, Medium, High, or Critical
- Each alert shows current usage, target, deviation percentage, and possible causes
- Manager investigates and selects cause: data error (auto-resolves on correction) or operational issue (records explanation and action)
- Manager resolves internally or escalates to COO with message
- COO acknowledges and responds back to Manager

F. Carbon Emissions Analysis

- System calculates CO_2 in kg from submitted data using standard emission factors
- CO_2 data visible only to Analyst and COO — not to Department Manager
- Analyst selects custom date range and department filter for analysis
- Single chart shows: blue area for consumption, cyan line for CO_2 on second Y axis, red dotted line for target
- Chart fills light blue below target and light red above target
- Department vs organization average chart highlights departments above and below internal average

G. Sustainability Report Generation & Approval

- Analyst writes full report freely in plain text editor — no guided boxes or labels
- Left panel shows read-only data reference: figures, calculations, and charts
- Analyst downloads PDF and submits to COO for approval
- COO approves or requests revision with comments
- Analyst edits directly based on COO comments and resubmits
- Both COO and Analyst view full report history with status and revision comments

H. Submissions Status Monitoring

- Analyst views submission status per department: Submitted or Pending
- Progress bar shows X of Y departments submitted
- Analyst sends reminder to pending departments
- Success banner shown when all submitted

I. Notifications

- Bell icon shows unread count — clicking opens full notification dropdown
- Each notification shows type, department, and timestamp with unread dot indicator

- Clicking navigates to relevant page and marks as read
- Notifications strictly role-filtered — no crossover between roles

J. Audit Trail

- Every important action logged with timestamp, actor, role, action, and department
- COO views and filters audit trail by date range, role, and department

K. Data Confidence Score

- System calculates confidence score for each submission as a percentage
- Above 75% green, 50–75% amber, below 50% red
- Expanding submission row shows factors: invoice verified, on time, pattern consistent, values match

Non-Functional Requirements

A. Performance

- Page load within 2 seconds, chart rendering within 1 second
- All numerical values displayed to maximum 2 decimal places
- System handles multiple concurrent users without degradation

B. Usability

- Role-based dashboards show only relevant information per actor
- Consistent button and status pill system across all pages
- All charts include titles, axis labels with units, legends, and hover tooltips
- Confirmation modals for all destructive actions
- Desktop-first design at 1440x1024

C. Security

- Secure login with role-based access control strictly enforced
- Only COO can change staff email addresses
- Password reset via OTP, sensitive data encrypted

D. Reliability

- 24/7 availability with data backup and recovery mechanisms
- Audit trail maintains complete tamper-proof logs
- High reliability for CO_2 calculations

E. Scalability

- Supports growing departments and staff
- Historical data archived without performance impact

F. Maintainability

- Modular architecture, standard UML practices followed
- Web-based with cloud deployment support

G. Compliance & Ethics

- Compliance-ready sustainability reports aligned with environmental standards
- Accurate and transparent representation of all consumption and CO_2 data
- No manipulation of sustainability metrics
- Compliant with data protection laws

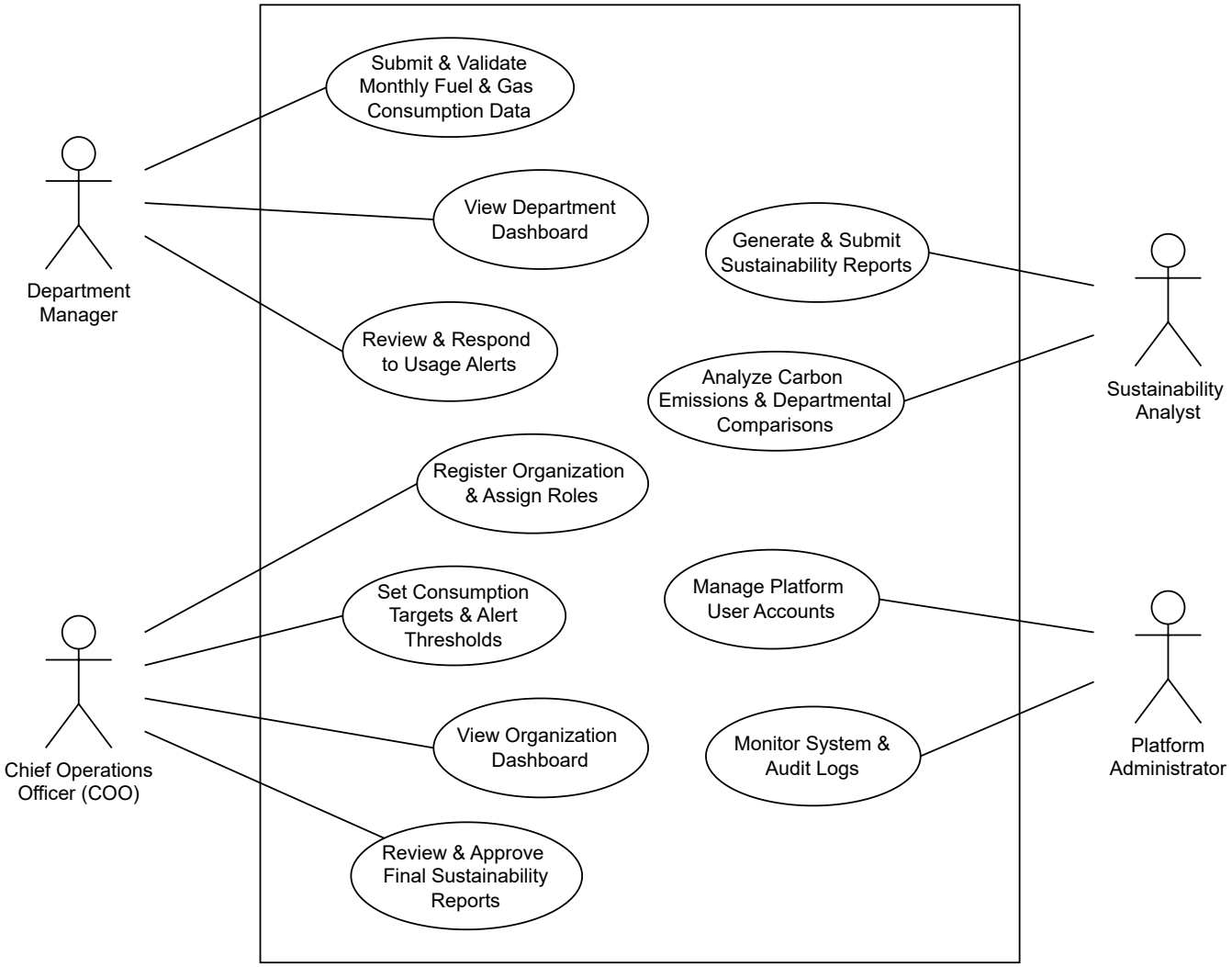
UML Diagrams

The system includes the following UML diagrams:

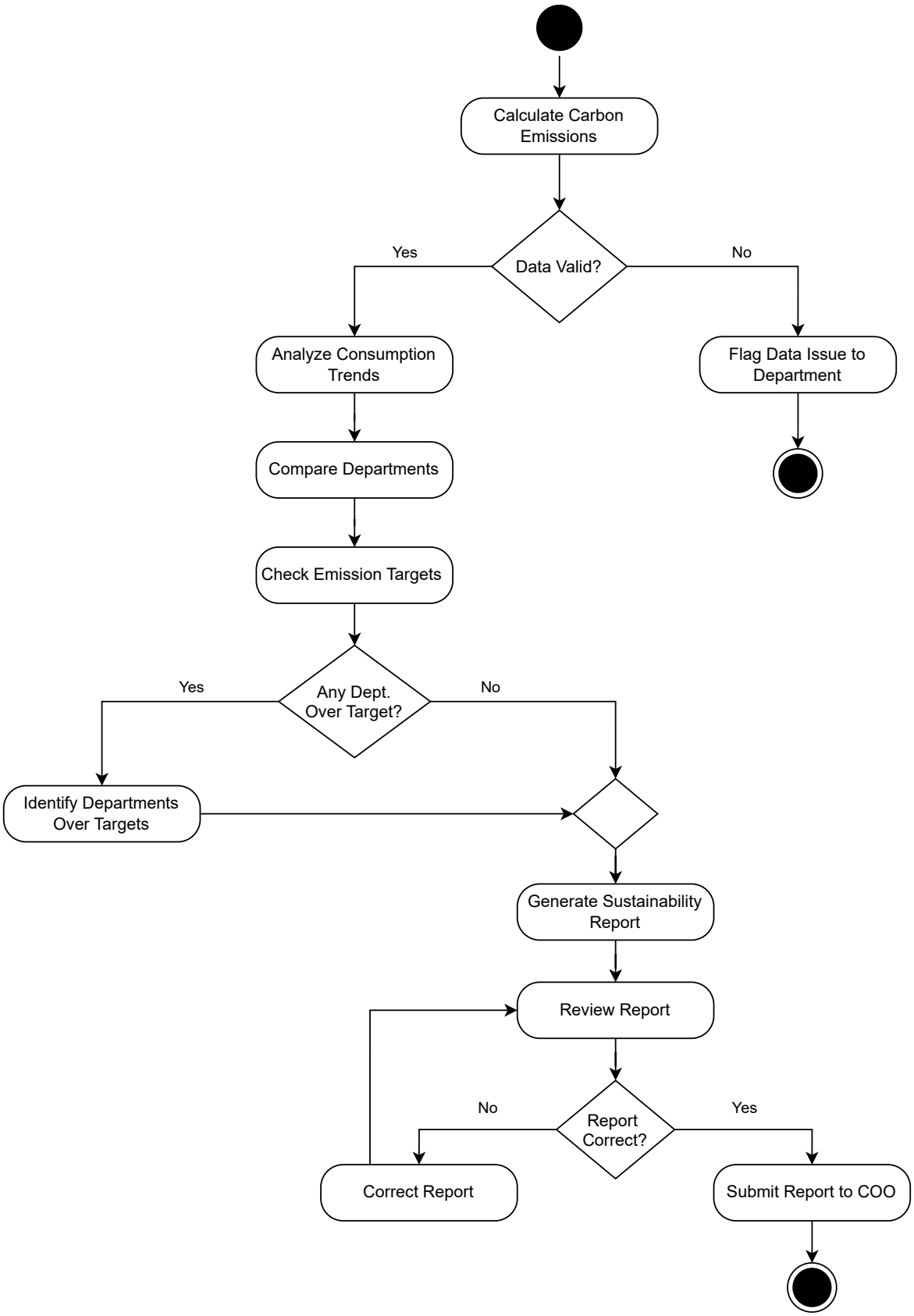
- Use Case Diagrams
- Activity Diagrams
- Sequence Diagrams

UML Diagrams are shown below:

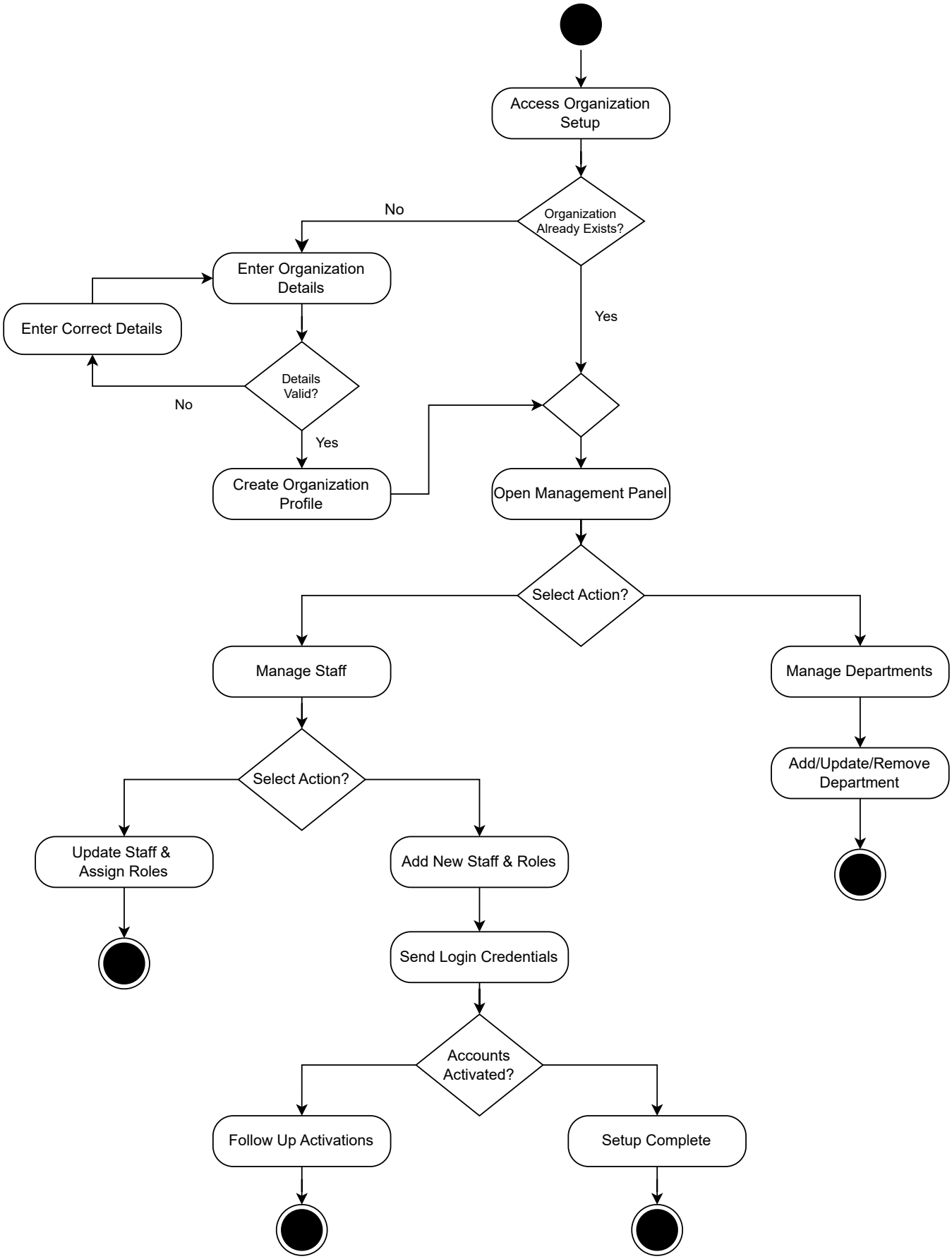
Resource Usage and Impact Analytics Platform



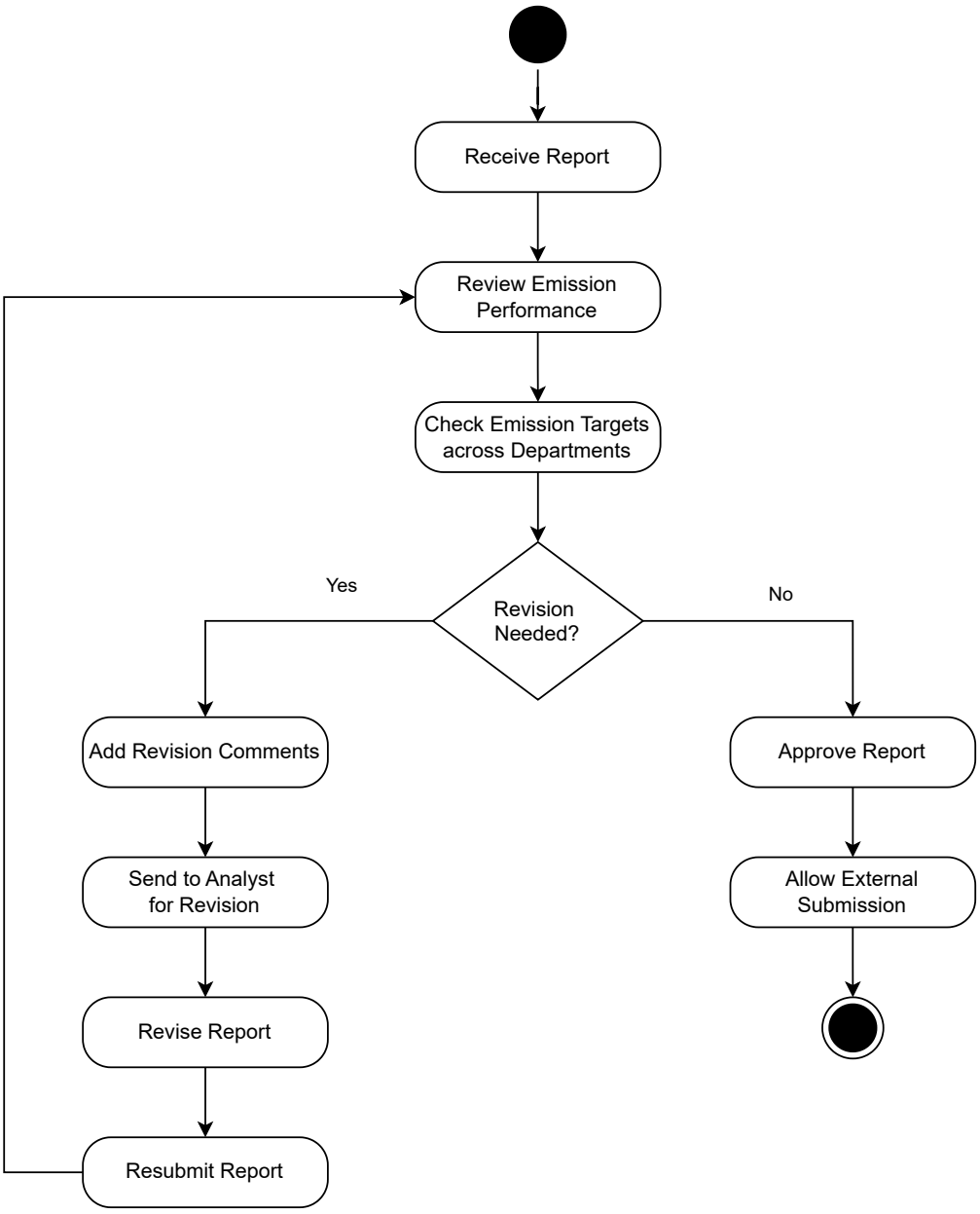
Analyze Consumption Trends & Carbon Emissions



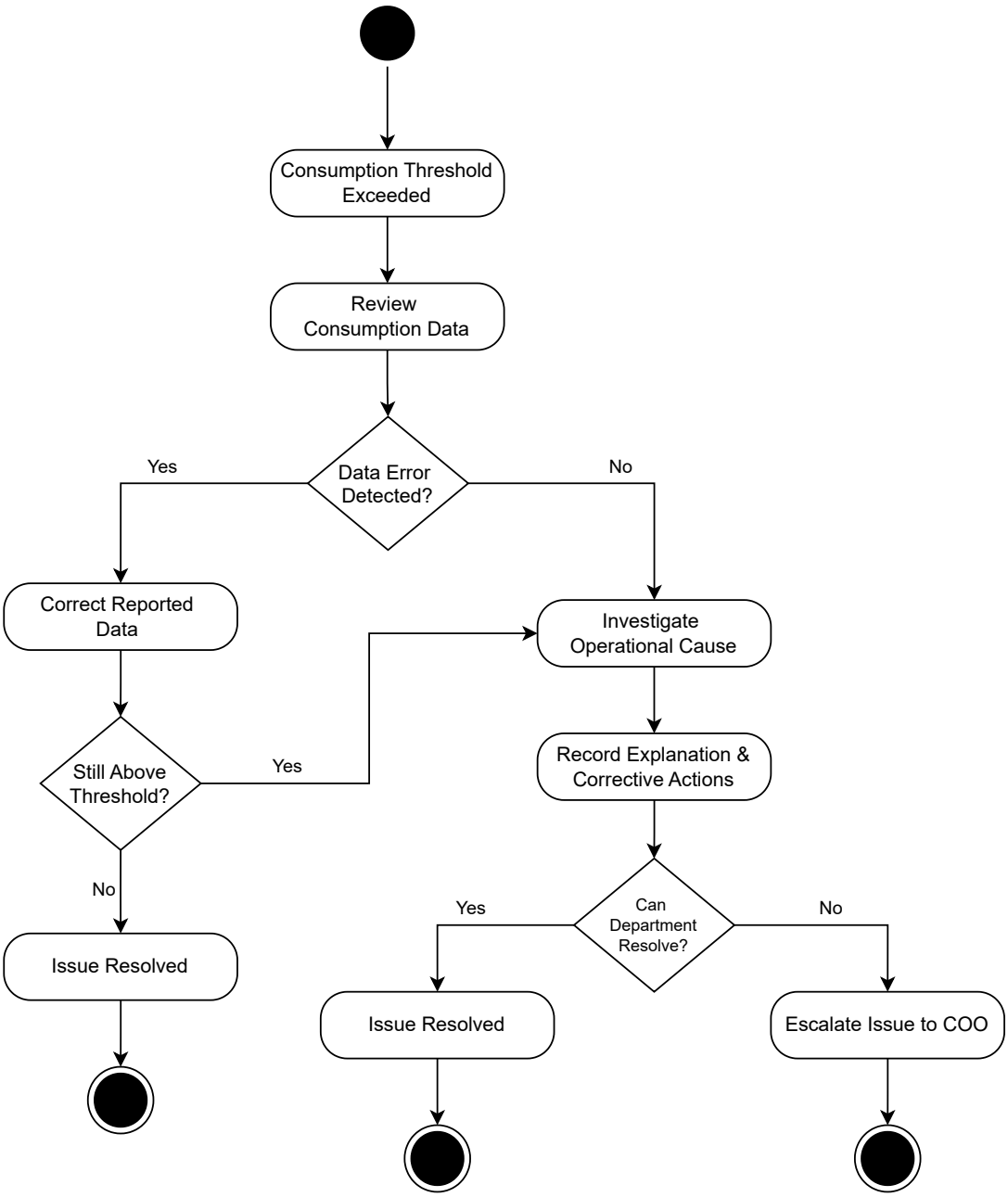
Organization Setup & Role Assignment



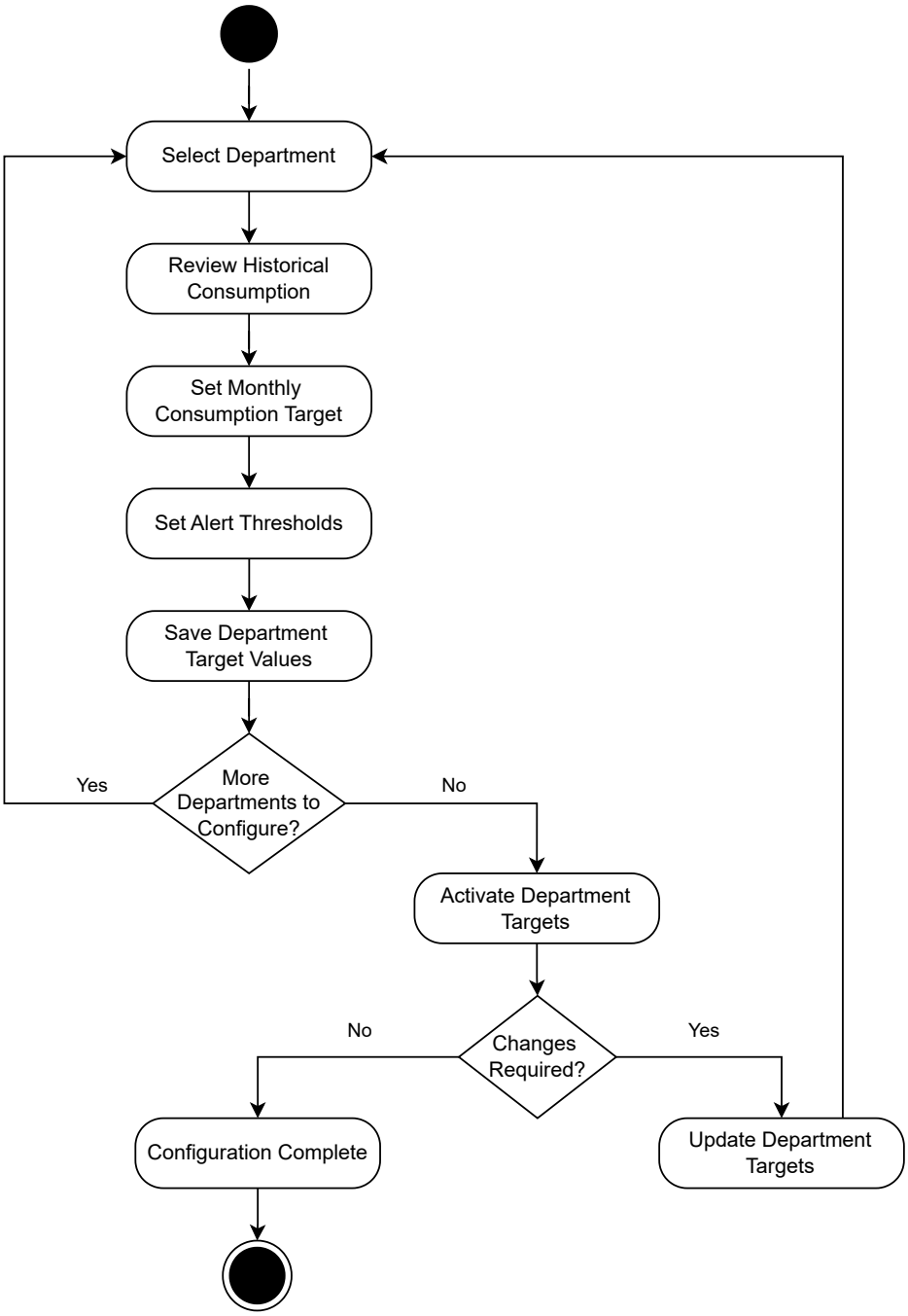
Review & Approve Sustainability Reports



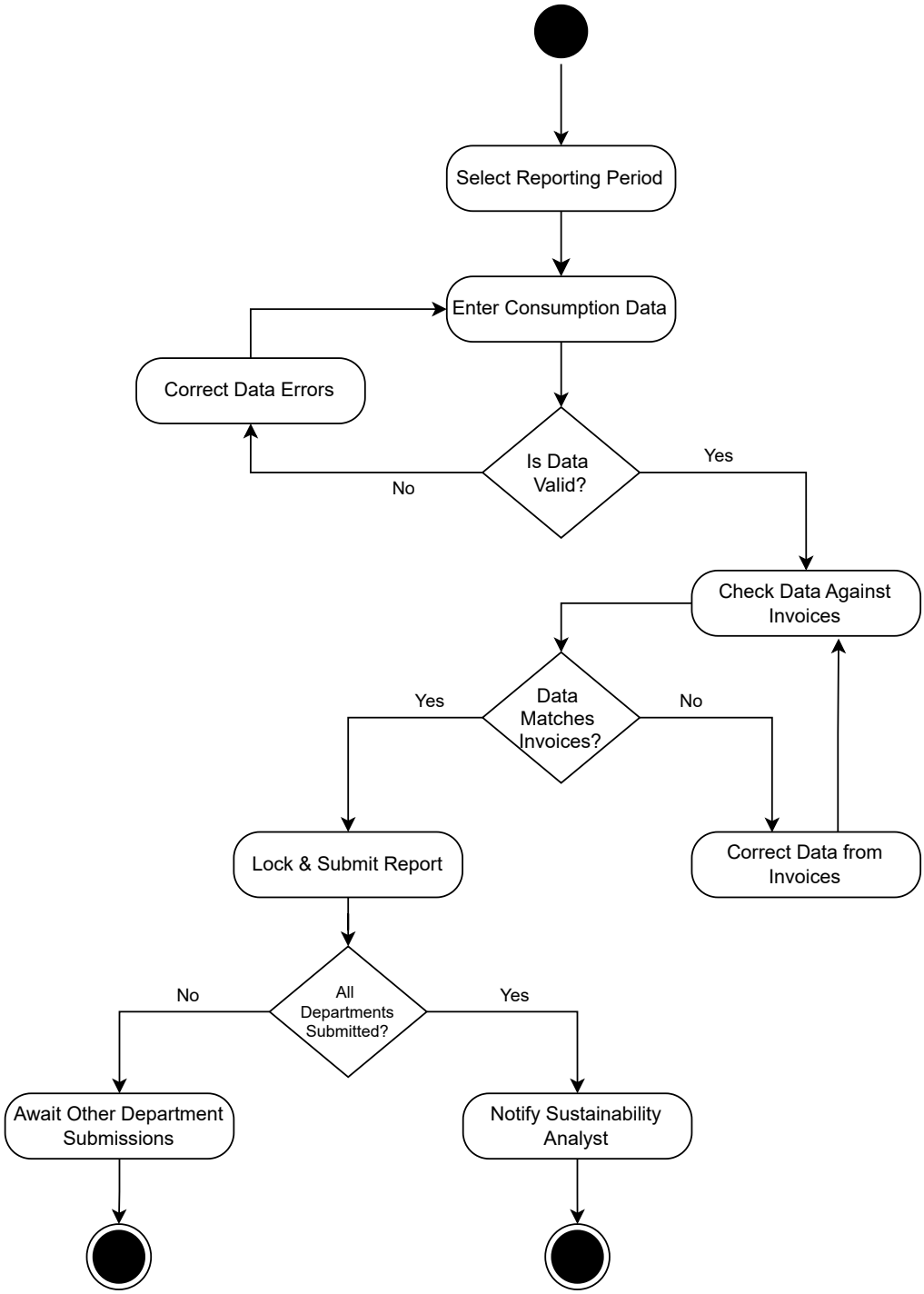
Review & Respond to Usage Alerts



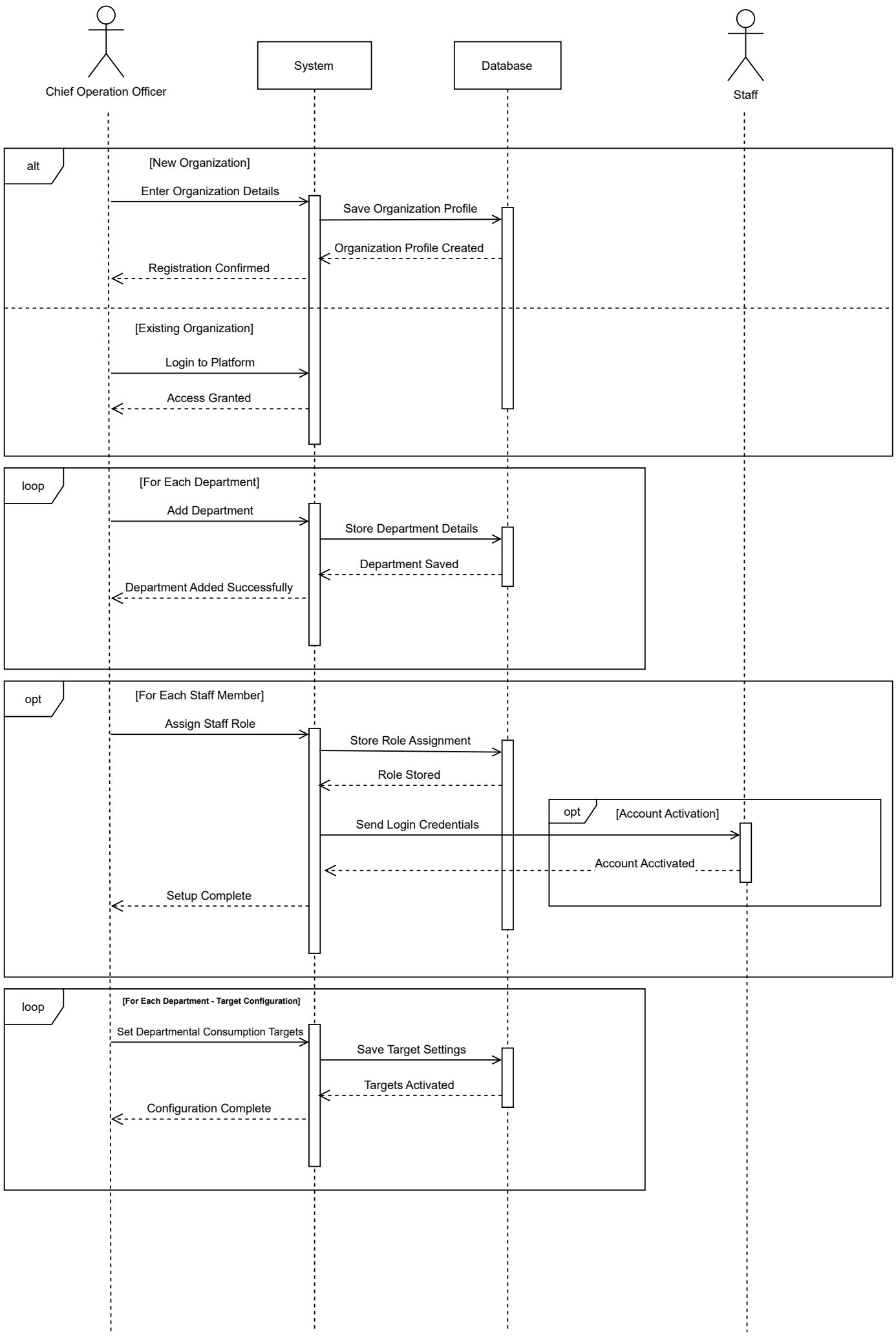
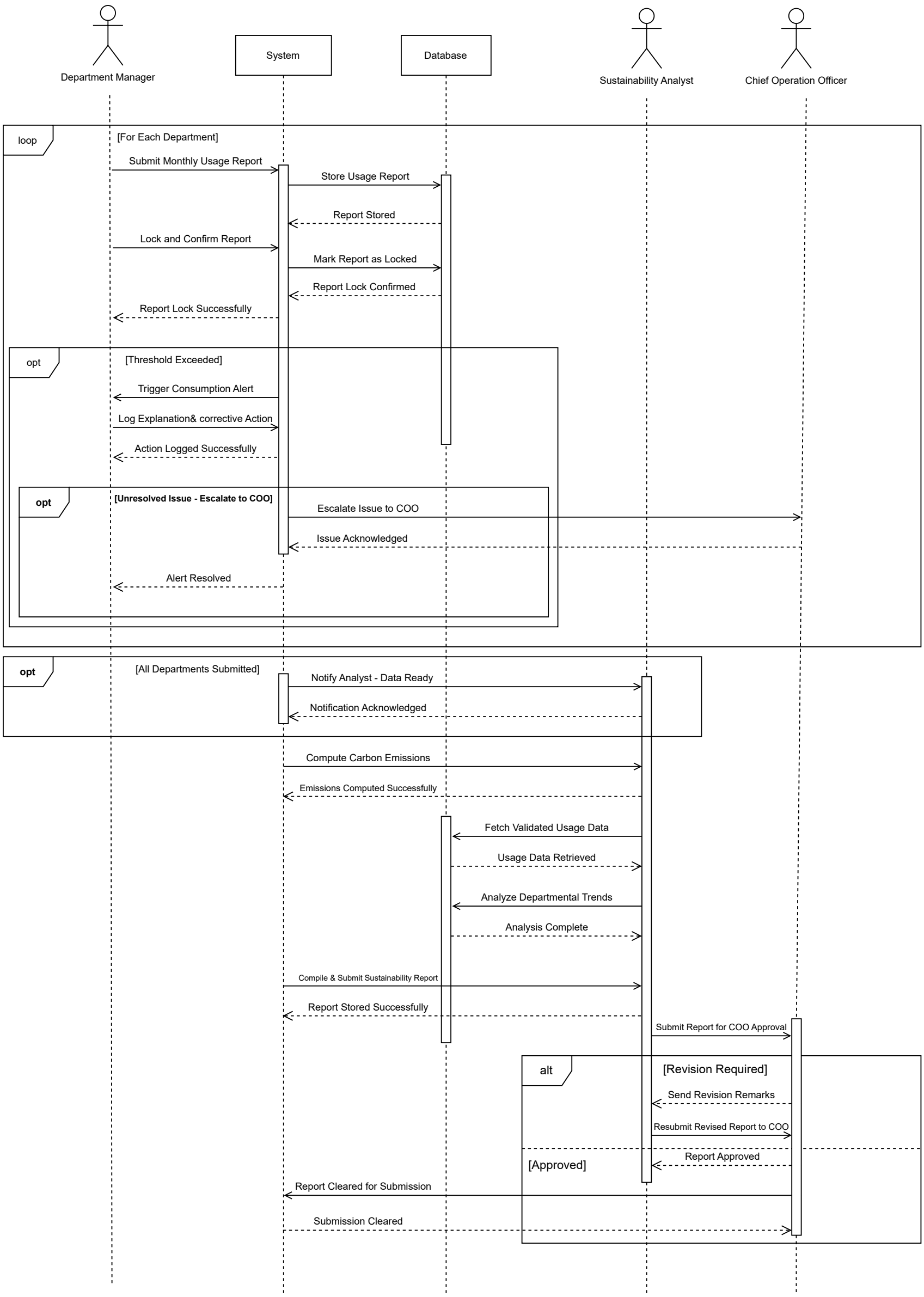
Set Consumption Targets & Alert Thresholds



Submit Monthly Resource Usage Data



Organization Setup and System Configuration



Data Submission,CO2 Analysis & Report Approval